

Deep Learning and Its Application

Week 9: Practicals

The goals of this practice are to have a clear understanding of some key concepts about reinforcement learning and Markov Decision Process. You need to research by reading recommended books or web materials before you answer the questions. You also need to exercise on answering the questions in E1.2

E1.1 Labwork:

- 1) Read Reinforcement Learning – Implement Grid World at <https://towardsdatascience.com/reinforcement-learning-implement-grid-world-from-scratch-c5963765ebff>
- 2) Download the code and create a Python file with the name of “grid-world” using PyCharm, cut and past the code into the grid-world
- 3) Run the rid-world
- 4) Understand next state is chosen
- 5) Understand how an action is determined
- 6) Look at the slides 21-26 of lecture Week 9, how Q-values are calculated
- 7) Change time-steps and episodes in the code and understand their impact on the game

E1.2 based on the game to define in your own words: (a) agent (b) environment (c) state (d) action (e) expected return (f) value iteration